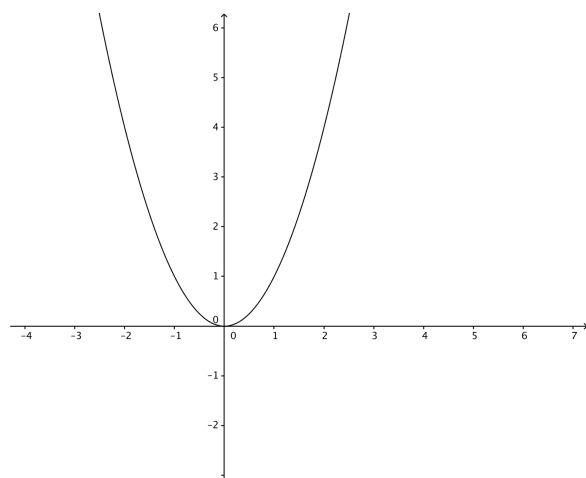


Problem 4

Problem 4

Problem 1 The graph of a function f is the parabola given in the figure below.



- (a) Find a formula for $f(x)$.
- (b) Suppose F_1 is an anti-derivatives of f satisfying $F_1(0) = 2$. Sketch and label the graph of F_1 .
Sketch the graphs of three more antiderivatives of f .
- (c) Using the expression you found in part (a) and your sketches in part (b), find the algebraic representation of F_1 .
- (d) Suppose we're given $h(x) = (1/3)x^3 + 17$, what is $h'(x)$?
- (e) What is the relationship between f , F_1 , h , and h' ?